

RESEARCH PROJECT 01
FEDERICO LOLLI AI RESEARCH

Symmetric Factorized Quadratic Adaptation

A Controlled Study of Nonlinear Low-Rank Adapters for GPT-2

Federico Lolli

xxlilloxx@gmail.com

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Abstract

Parameter-efficient fine-tuning adapts pretrained models through small modules while leaving the backbone frozen. This study asks whether a low-rank adapter gains useful capacity by replacing the linear LORA correction with an explicit quadratic function of projected activations. The proposed SYMMETRIC adapter projects an activation to a low-dimensional space, squares its coordinates element by element, and maps them back; it has the same trainable matrix shapes and parameter count as matched LORA. An independent 24-run WikiText-2 confirmation covers ranks 1, 2, 4, and 8, three seeds, and 5,000 optimizer steps. SYMMETRIC obtains lower mean final validation and held-out test loss at every tested rank, with all twelve paired test comparisons favouring it. A six-run AG News classification study provides the counter-result: LORA achieves higher mean accuracy and macro-F1. A single-seed scaling ablation and fresh convergence trajectories qualify the rank result. Analytical and empirical studies show that LORA has an input-independent Jacobian and zero adapter-only Hessian, whereas SYMMETRIC has an activation-dependent local Jacobian and explicit second-order response. A matched rank-4 benchmark finds nearly identical latency and a 1.25 MiB increase in peak allocated VRAM. The evidence supports quadratic activation-space adaptation as a viable, task-dependent design point rather than a universally superior replacement for linear low-rank adaptation.

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1 Introduction

Full fine-tuning updates every model weight, whereas parameter-efficient fine-tuning (PEFT) freezes most pretrained parameters and learns a small correction. This reduces task-specific state and makes the correction’s functional form a meaningful experimental variable. LORA [1] is a natural baseline: it is simple, parameter efficient, and interpretable as one constant low-rank weight update. Its adapter branch is nevertheless linear in the incoming activation and cannot itself express second-order feature interactions.

This study evaluates a small change in function class. The incoming activation is projected to r coordinates, those coordinates are squared element by element, and the result is projected back. This preserves the trainable shapes of LORA while making the correction quadratic and activation dependent. A controlled comparison is necessary because rank, parameter count, placement, scaling, training duration, seed, and task can all alter the observed ordering.

1.1 Research question

Can a low-rank adapter gain useful expressive capacity by replacing the purely linear LORA correction with an explicit quadratic function of projected activations, while retaining a comparable parameter budget and practical computational cost? The question matters because adapters of identical size can implement different local transformations.

1.2 Contributions

This work provides (i) a precise symmetric factorized quadratic activation-space adapter; (ii) controlled comparisons with LORA under matched budgets; (iii) rank, scaling, convergence, placement, and downstream analyses; (iv) local-Jacobian, effective-rank, and adapter-Hessian analyses; (v) a matched cost benchmark; and (vi) conservative positioning relative to verified related methods. These are empirical and analytical contributions, not a claim of universal novelty or superiority. Sections 2–4 introduce the context, method, and protocols; Sections 5–8 present results, mechanism analysis, and cost; Sections 9–11 discuss scope and conclusions.

2 Related work and positioning

Table 1 compares mechanisms rather than incompatible benchmark scores. Only LORA is reproduced as a direct baseline. LORA learns a linear update [1]. DoRA separates weight magnitude and direction [2]; MoRA uses a square matrix and fixed transforms [3]; HiRA forms a Hadamard-product update [4]; and LoRAN transforms a low-rank weight update nonlinearly [5]. These operate in weight or parameter space rather than by squaring the current activation.

2.1 PERA

PERA expands low-rank factors polynomially before composition [6]. This study follows $x \rightarrow xU \rightarrow (xU) \odot (xU) \rightarrow P$. Both investigate higher-order structure, but the operation acts on different objects. PERA is not an experimental baseline here.

Table 1: Mathematical and architectural positioning. External methods were not experimentally reproduced.

Method	Core mechanism	Relation to this study
LoRA	Low-rank linear correction	Direct baseline
DoRA	Weight magnitude/direction decomposition	Different weight parameterization
MoRA	Square trainable matrix with fixed transforms	Weight-space high-rank strategy
HiRA	Hadamard-product high-rank update	Weight-space higher-rank adaptation
LoRAN	Nonlinear low-rank weight update	Nonlinear weight-space method
PERA	Polynomial expansion of low-rank factors	Higher-order parameter/factor-space method
QuadraNet V2	Factorized quadratic neural transformation	Related quadratic-form precedent
This study	Squared projected activation	Activation-dependent quadratic correction

2.2 QuadraNet V2

QuadraNet V2 is a verified precedent for factorized quadratic transformations [7]. For one output, the present adapter is xQ_jx^T . Across a multi-output layer, all Q_j share directions in U and differ through coefficients in P ; single-output similarity does not establish full architectural equivalence.

2.3 Position of the proposed method

The defining feature is the element-wise square after the activation projection and before the output projection. This yields an input-dependent local transformation at matched LoRA size. Figure 9 later summarizes where higher-order operations enter the compared methods.

3 Method

3.1 Frozen projection and parallel correction

For row activation $x \in \mathbb{R}^{1 \times d}$, frozen $W \in \mathbb{R}^{d \times d_{out}}$, and frozen bias b ,

$$y_{base} = xW + b. \quad (1)$$

The adapter is a parallel trainable branch. Matched LoRA and SYMMETRIC corrections are

$$\Delta y_{LoRA} = \frac{\alpha}{r}(xA)B, \quad (2)$$

$$\Delta y_{Sym} = \frac{\alpha}{r}((xU) \odot (xU))P, \quad (3)$$

where $A, U \in \mathbb{R}^{d \times r}$ and $B, P \in \mathbb{R}^{r \times d_{out}}$. Both contain $r(d + d_{out})$ parameters, or $2dr$ for square GPT-2 c_proj. LoRA applies two linear projections; SYMMETRIC forms $z = xU$, computes $z \odot z$, then applies P .

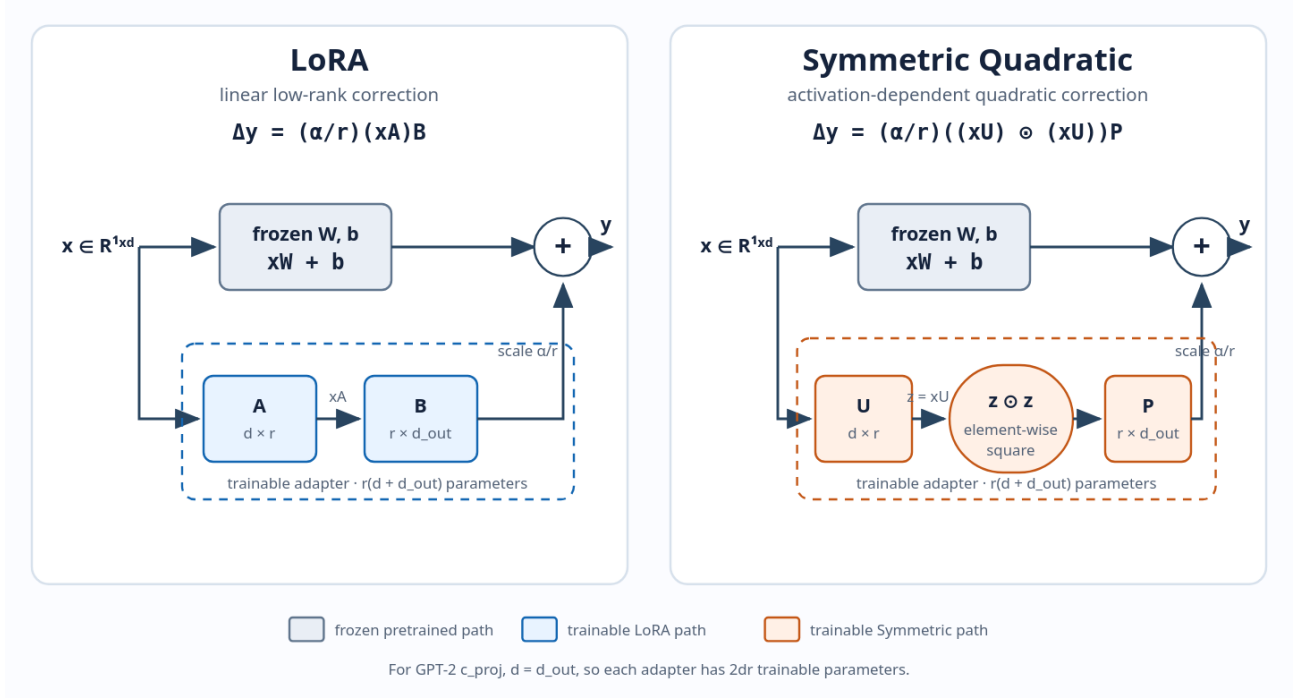


Figure 1: Architecture of matched LoRA and SYMMETRIC adapters. Both add a low-rank trainable branch beside a frozen projection. LoRA is linear; SYMMETRIC inserts an element-wise square before the output projection.

3.2 Quadratic structure

For output j ,

$$\Delta y_j = \frac{\alpha}{r} \sum_k P_{kj} (x u_k)^2 = x Q_j x^T, \quad (4)$$

$$Q_j = \frac{\alpha}{r} U \text{diag}(P_{:,j}) U^T. \quad (5)$$

Q_j is symmetric and $\text{rank}(Q_j) \leq r$. Directions in U are shared across outputs; signed coefficients in P mean Q_j need not be positive semidefinite. The identity $(ax_1 + bx_2)^2 = a^2x_1^2 + 2abx_1x_2 + b^2x_2^2$ shows that squared projections create cross terms. Output factors are zero initialized. The main study uses $\alpha = 4$, giving $\alpha/r = 4, 2, 1, 0.5$ at ranks 1, 2, 4, 8 and motivating Section 6.

3.3 GPT-2 insertion point

The controlled adapter receives the concatenated attention output entering zero-based block 0 `attn.c_proj`. Frozen projection and adapter outputs are summed before attention residual dropout and residual addition (Figure 2).

Final-study adapter location in a zero-based GPT-2 block

Block 0 is the first Transformer block (zero-based indexing). Frozen modules are gray; trainable adapter is blue.

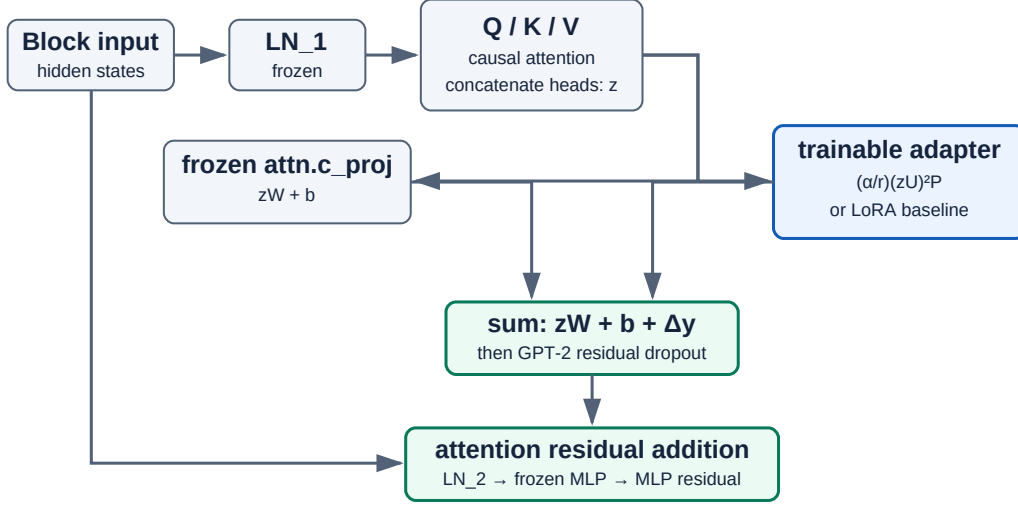


Figure 2: Controlled GPT-2 placement. The trainable adapter and frozen block-0 `attn.c_proj` share an input; their outputs are summed before residual dropout and the attention residual addition. Block numbering is zero based.

4 Experimental design

Historical studies motivated controlled questions but are not pooled with them. Table 2 records each campaign’s role.

Table 2: Completed experimental campaigns.

Campaign	Design	Role
Adapter families	Linear, quadratic, interaction, symmetric, mixed variants	Exploration
Depth/placement	Attention/MLP screens and 12-block scan	Placement
Fixed-budget multi-layer	6,144/12,288 parameters across layers	Depth/local rank
AG News	Two methods, three seeds, 500 steps	Classification
Rank screening	Two methods, four ranks, three seeds, 1,000 steps	Screening
Rank confirmation	24 fresh runs, 5,000 steps	Central LM evidence
Scaling ablation	$\alpha = 4$ versus $\alpha/r = 1$, seed 42	Scale check
Mechanism/cost	Stored checkpoints; matched rank-4 benchmark	Structure/cost

Table 3 consolidates the two principal controlled protocols. The independent 1,000-step rank screen is a separate campaign and is not the first segment of the 5,000-step confirmation trajectories.

Table 3: Controlled experimental protocols. Fields are taken from the recorded publication-safe configurations.

Setting	WikiText-2 confirmation	AG News
Backbone	GPT-2 Small, frozen	GPT-2 Small, frozen
Placement	Block 0 <code>attn.c_proj</code>	Block 0 <code>attn.c_proj</code>
Objective	Causal language modelling	Classification
Sequence length	128 tokens	128 tokens
Training data	36,718 packed blocks	4,096 examples
Validation	3,760 packed blocks; eight-block evaluation limit	1,000-example stratified subset
Test	Held-out WikiText-2 test; 2,213 evaluated blocks	Original AG News test split
Methods	LoRA, SYMMETRIC	LoRA, SYMMETRIC
Rank(s)	1, 2, 4, 8	4
Seeds	42, 123, 456	42, 123, 456
Optimizer / learning rate	AdamW / 3×10^{-4}	AdamW / 3×10^{-4}
Batch / accumulation	1 / 4	1 / 4
Gradient clipping	1.0	1.0
Scheduler	None	None
Adapter scale	$\alpha = 4$	$\alpha = 4$
Training steps	5,000	500
Trainable parameters	Rank dependent; matched by method	9,216 including classifier

4.1 Historical exploration

Early adapter-family studies, 500/1,000/2,000-step durations, attention and MLP screens, a 12-block attention scan, and fixed-budget multilayer designs established trainability and motivated later controls. Their nonuniform protocols remain separate; Appendix B summarizes verified values.

4.2 Controlled WikiText-2 protocol

GPT-2 Small is frozen. WikiText-2 is packed into consecutive 128-token blocks for causal cross-entropy. Runs use block 0 `attn.c_proj`, batch 1, accumulation 4, AdamW at 3×10^{-4} , clipping 1.0, no scheduler, $\alpha = 4$, ranks 1/2/4/8, seeds 42/123/456, and 5,000 steps. The 1,000-step screen is independent. Best recorded validation, exact final validation, and final-checkpoint held-out test are distinct; test data select nothing.

4.3 Controlled AG News protocol

The same frozen backbone and placement are used with a classification head, 4,096 training examples, a fixed 1,000-example stratified validation subset, the original test split, rank 4, three seeds, and 500 steps. Each method has 9,216 trainable parameters including the head.

4.4 Scaling ablation

A seed-42, 1,000-step ablation compares fixed $\alpha = 4$ with constant $\alpha/r = 1$ ($\alpha = r$). It is exploratory and separate from the main confirmation.

4.5 Mechanism and cost protocols

Final checkpoints are reused for spectra, output covariance, local Jacobians, and adapter-only Hessians on fixed held-out activations. Analytical derivatives are checked against double-precision autograd.

The cost benchmark uses rank 4, batch 1, sequence length 128, 10 warm-up and 30 measured iterations. No full-block Hessian was completed.

5 Controlled results

5.1 WikiText-2 confirmation

Table 4 reports final-checkpoint means and sample SD. Lower is better. Paired test differences

Table 4: Final WikiText-2 loss over three seeds.

Rank	LoRA val	SYMMETRIC val	LoRA test	SYMMETRIC test
1	3.5734 ± 0.0051	3.5314 ± 0.0207	3.9256 ± 0.0009	3.9131 ± 0.0042
2	3.4773 ± 0.0057	3.4340 ± 0.0249	3.8492 ± 0.0025	3.8305 ± 0.0144
4	3.3483 ± 0.0093	3.3101 ± 0.0011	3.7656 ± 0.0054	3.7429 ± 0.0024
8	3.3054 ± 0.0117	3.2446 ± 0.0096	3.7319 ± 0.0035	3.6938 ± 0.0044

(SYMMETRIC minus LoRA) are -0.0125 ± 0.0051 , -0.0187 ± 0.0119 , -0.0227 ± 0.0035 , and -0.0381 ± 0.0072 at ranks 1,2,4,8; all 12 pairs favour SYMMETRIC. Figure 3 shows individual seeds and SD. This result is specific to the tested task, placement, schedule, and scale.

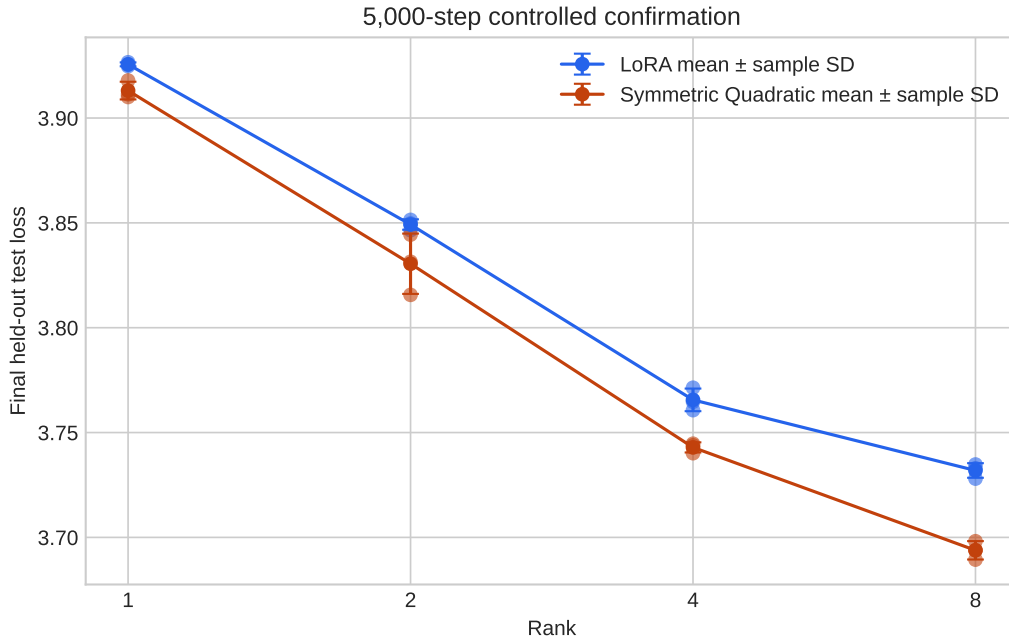
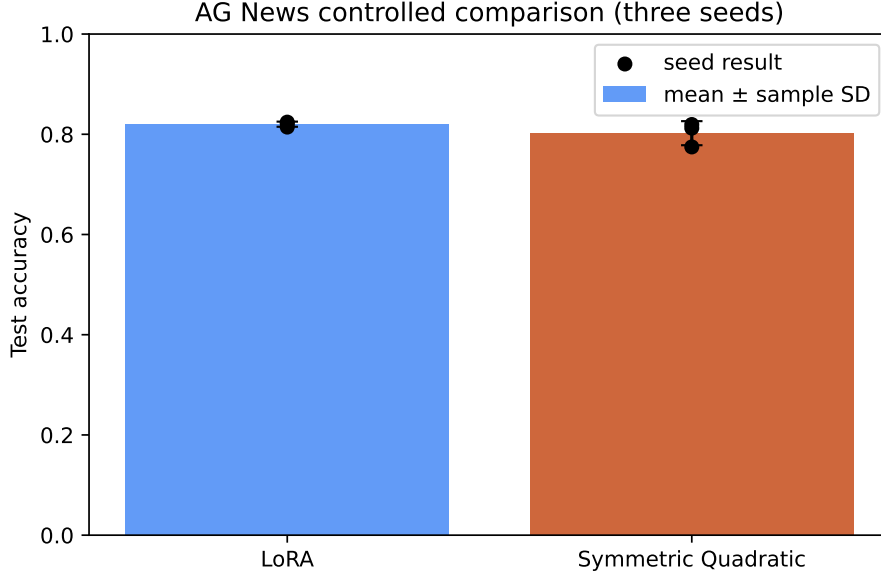


Figure 3: Held-out WikiText-2 test loss after 5,000 steps at ranks 1,2,4,8. Lower is better. Points are seeds; markers/error bars are mean/sample SD. SYMMETRIC is lower on average at every rank in this protocol.

Table 5: AG News held-out results, mean $\pm$ sample SD.

Method	Accuracy	Macro-F1	Test loss
LoRA	0.8201 ± 0.0051	0.8156 ± 0.0050	0.5445 ± 0.0542
SYMMETRIC	0.8021 ± 0.0240	0.7929 ± 0.0315	0.6847 ± 0.1919

**Figure 4:** AG News held-out accuracy. Bars/error bars are mean/sample SD; black points show all seeds. LoRA is higher on average.

5.2 AG News

LoRA is better on average for all three measures. Figure 4 exposes the larger SYMMETRIC seed variability, demonstrating that the WikiText-2 ordering does not transfer automatically.

6 Scaling and convergence

6.1 Constant-scale ablation

Table 6: Seed-42 validation loss after 1,000 steps; $\Delta = \text{SYMMETRIC} - \text{LoRA}$.

Rank	$\alpha = 4$			$\alpha/r = 1$		
	LoRA	SYMMETRIC	Δ	LoRA	SYMMETRIC	Δ
1	3.6171	3.6614	+0.0443	3.6301	3.6711	+0.0410
2	3.5861	3.5561	-.0300	3.5984	3.5693	-.0292
4	3.5530	3.4780	-.0750	3.5530	3.4780	-.0750
8	3.5492	3.4315	-.1177	3.5056	3.4081	-.0975

The sign pattern persists for seed 42 while magnitude changes (Figure 5); one seed prevents broader inference.

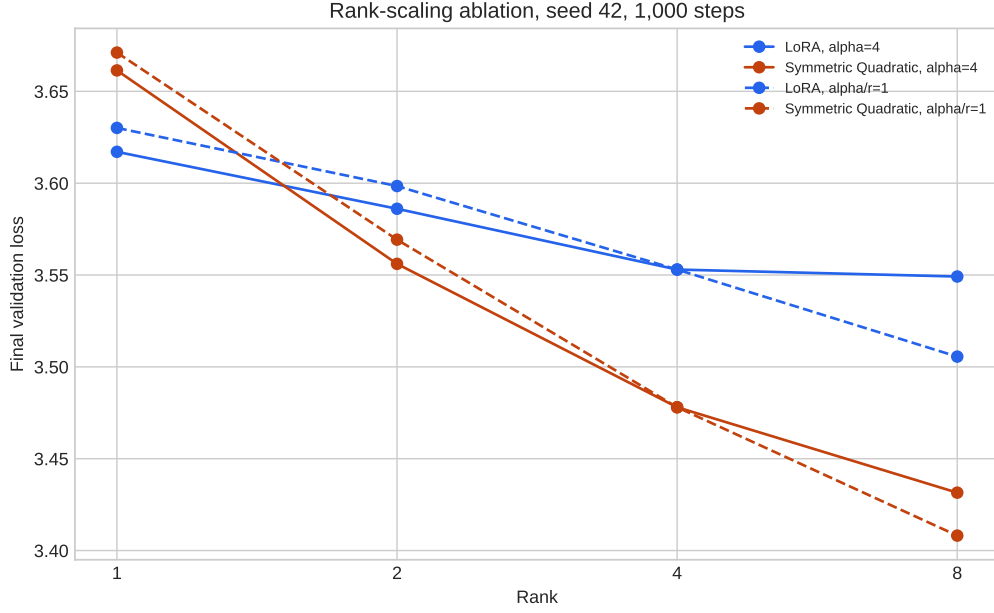


Figure 5: Single-seed scaling ablation. Fixed α changes effective scale with rank; the comparison fixes $\alpha/r = 1$.

6.2 Fresh confirmation trajectories

Table 7: Mean paired validation difference SYMMETRIC – LoRA.

Rank	1,000	2,000	3,000	4,000	5,000
1	+.0280	+.0096	-.0043	-.0265	-.0419
2	-.0367	-.0432	-.0319	-.0311	-.0434
4	-.0707	-.0631	-.0803	-.0635	-.0382
8	-.1247	-.1156	-.0947	-.0781	-.0608

Negative favours SYMMETRIC. Rank 1 crosses between 2,000 and 3,000 recorded steps; ranks 2–8 favour SYMMETRIC at every listed point. The independent screen is not joined to these curves. Figure 6 shows rank 1 and SD; no asymptotic claim is made.

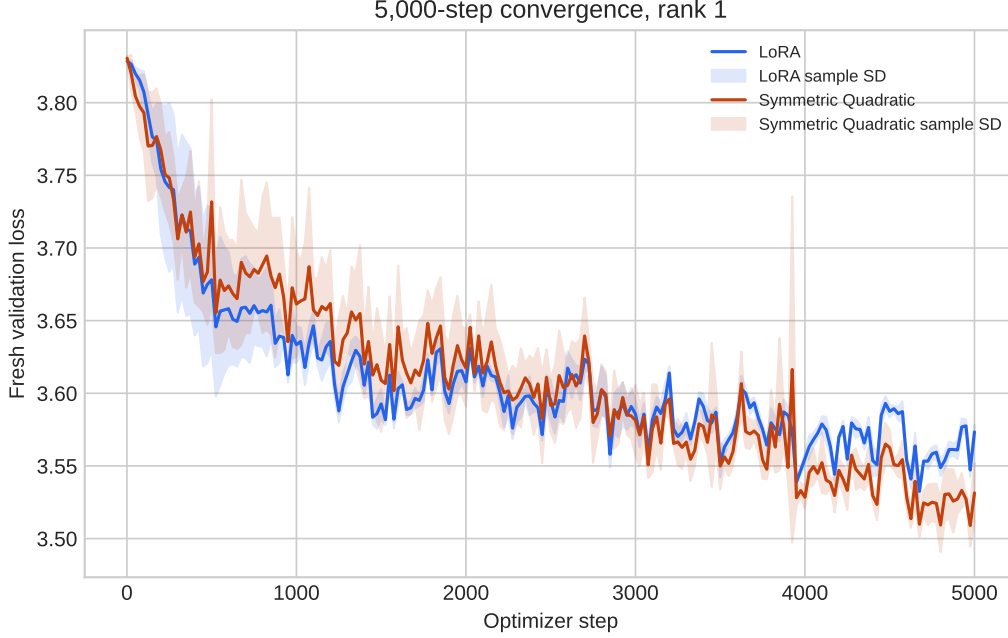


Figure 6: Fresh rank-1 validation trajectories in the 5,000-step confirmation. Curves are means; bands are sample SD.

7 Mechanism analysis

These analyses characterize structure but do not establish why performance differs.

7.1 Local Jacobians

$$J_{\text{LoRA}} = \frac{\alpha}{r} AB, \quad (6)$$

$$J_{\text{Sym}}(x) = 2\frac{\alpha}{r} U \text{diag}(xU)P. \quad (7)$$

LoRA is input independent and represents one fixed local update. SYMMETRIC changes with x and cannot generally be reduced to one constant ΔW . Double-precision tests match these expressions to autograd.

7.2 Effective rank

Entropy effective rank is $\exp(-\sum_i p_i \log p_i)$ for normalized singular values p_i . Table 8 reports distinct objects. The SYMMETRIC local-Jacobian rank is not a fixed weight-update rank directly equivalent to LoRA (Figure 7).

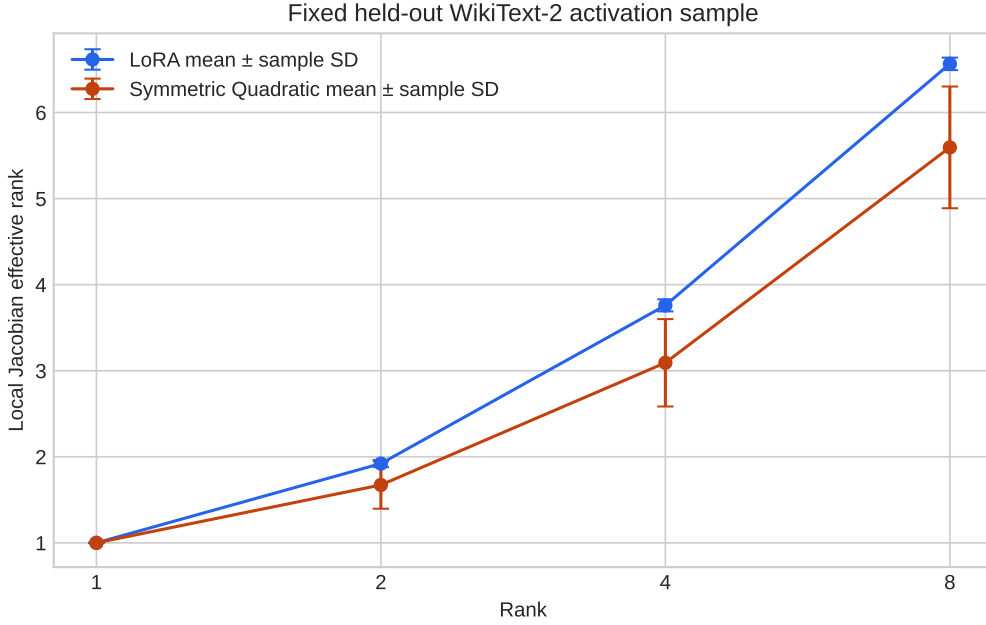
7.3 Second-order structure

For fixed linear output reduction c ,

$$H = 2\frac{\alpha}{r} U \text{diag}(Pc)U^T. \quad (8)$$

Table 8: Mean entropy effective ranks.

Rank	LoRA update	SYMMETRIC Jacobian	U	P	Output cov.
1	1.00	1.00	1.00	1.00	1.00
2	1.92	1.67	2.00	1.98	1.92
4	3.76	3.09	3.99	3.89	3.00
8	6.57	5.59	7.96	7.28	3.59

**Figure 7:** Effective rank of the LoRA update and SYMMETRIC local Jacobian. These measure related local complexity but are not identical weight-space objects.

LoRA has zero adapter-only Hessian; SYMMETRIC is nonzero. Mean SYMMETRIC Frobenius norms are 155.15, 92.13, 52.68, 17.61 at ranks 1,2,4,8. The decrease combines learned factors and scale and is not evidence of weaker useful interactions. For fixed c , the Hessian is input independent, so repeated tokens are not independent observations. Autograd verifies Equation 8; a full-block Hessian was not evaluated.

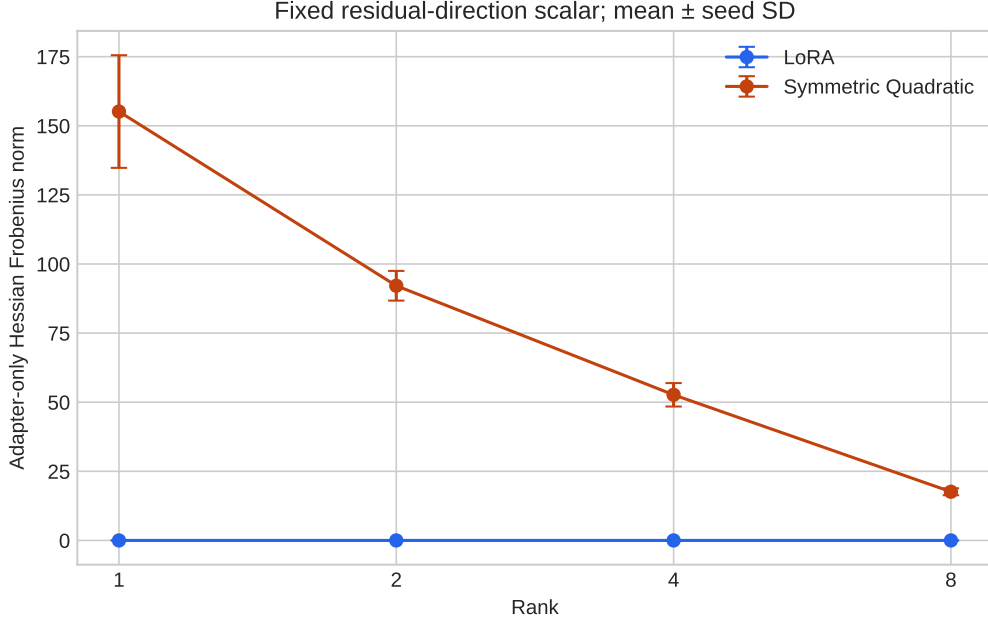


Figure 8: Adapter-only second-order response. LoRA is exactly zero; SYMMETRIC has an explicit nonzero response. This does not prove a causal performance mechanism.

8 Computational cost

Both rank-4 adapters have 6,144 parameters. Table 9 shows nearly identical measured cost. Relative

Table 9: Matched rank-4 benchmark: batch 1, length 128, 10 warm-up and 30 measured iterations.

Method	Forward (ms)	Backward (ms)	VRAM (MiB)	Tokens/s
LoRA	12.157 ± 0.243	14.805 ± 0.236	429.05	10,529
SYMMETRIC	12.184 ± 0.236	14.766 ± 0.188	430.31	10,505

differences are approximately +0.22% forward latency, −0.27% backward latency, +1.25 MiB VRAM, and −0.22% throughput. The result applies only to this implementation and hardware configuration.

9 Discussion

The controlled experiments reveal a more nuanced picture than a simple ranking between LoRA and SYMMETRIC. The central WikiText-2 confirmation provides consistent evidence in favour of the quadratic adapter under the tested language-modelling protocol, whereas the AG News classification experiment reverses that ordering. The main interpretation is therefore not that one adapter dominates the other, but that the functional form of a low-rank correction can materially affect behaviour and that this effect depends on task, rank, scaling policy, and optimization regime.

9.1 WikiText-2 result and rank dependence

The strongest evidence comes from the independent 5,000-step WikiText-2 confirmation campaign. Across ranks 1, 2, 4, and 8, SYMMETRIC achieves lower mean final validation loss and lower mean held-out test loss than the matched LORA baseline. All twelve paired seed-rank held-out comparisons also show lower loss for SYMMETRIC under this protocol.

The mean held-out test difference increases across the tested ranks, from approximately -0.0125 at rank 1 to -0.0381 at rank 8. This shows that the observed difference is not confined to a single rank. Nevertheless, the present experiments do not establish why the magnitude changes with rank. Under the primary protocol, increasing rank changes both adapter capacity and the effective scale α/r , so rank alone cannot be treated as the sole causal variable.

The appropriate interpretation is configuration specific: the SYMMETRIC functional form is useful in the tested GPT-2 Small, WikiText-2, block-0 `attn.c_proj` setting. The experiment does not establish that quadratic adaptation must outperform linear adaptation for language modelling in general.

9.2 Optimization dynamics

The convergence analysis shows that training duration can materially change the comparison. Rank 1 is particularly informative. LORA has lower mean validation loss at the earlier recorded checkpoints, but the ordering changes between the 2,000- and 3,000-step evaluations. SYMMETRIC subsequently maintains the lower mean validation loss through the final 5,000-step checkpoint. A comparison terminated earlier could therefore have produced a different conclusion.

Ranks 2, 4, and 8 follow a different trajectory. SYMMETRIC already has lower mean validation loss at the recorded 1,000-step point. At ranks 4 and 8, however, the magnitude of the difference narrows later in training. These trajectories show that adapter comparisons should not be interpreted independently of optimization duration. They do not establish asymptotic behaviour, because the study observes only the finite schedules that were actually executed.

9.3 Scaling and rank

The main rank experiment fixes α at 4, so α/r decreases as rank increases. The constant-scale ablation tests whether this scaling rule alone accounts for the observed rank pattern. With α/r fixed to 1, the qualitative 1,000-step pattern remains unchanged for seed 42: LORA performs better at rank 1, while SYMMETRIC performs better at ranks 2, 4, and 8.

The magnitude does change. At rank 8, for example, the SYMMETRIC-minus-LORA validation-loss difference changes from -0.1177 under the primary scaling policy to -0.0975 under constant effective scale. This indicates that scaling contributes to the magnitude of the observed difference but does not, for the tested seed, fully explain the rank-dependent sign pattern. Because the constant-scale experiment was performed for only one seed, it cannot establish a general rank-scaling relationship.

9.4 Task dependence and AG News

AG News provides an important counterexample to the WikiText-2 result. Under the matched classification protocol, LORA achieves higher mean test accuracy, higher macro-F1, and lower test loss. SYMMETRIC also shows substantially greater seed-to-seed variability.

The two settings differ in several respects. WikiText-2 is a causal language-modelling task, whereas AG News is classification and introduces a trainable classification head. The datasets, objectives, and training budgets also differ. The present experiments do not isolate which factor causes the reversed ordering. The defensible conclusion is therefore that the relative benefit of the quadratic correction is task and protocol dependent. This negative counter-result is scientifically important because it prevents the positive WikiText-2 result from being interpreted as evidence of universal superiority.

9.5 Structural interpretation

The derivative analyses establish that LORA and SYMMETRIC belong to different local function classes. LORA is linear with respect to the adapter input. Its Jacobian is input independent and its adapter-only Hessian is exactly zero. Its correction can therefore be represented by a single fixed low-rank update. SYMMETRIC has an input-dependent local Jacobian and an explicit non-zero adapter-only second derivative. Its effective local transformation changes with the incoming activation and cannot, in general, be reduced to one constant ΔW .

The effective rank of the LORA update, the effective rank of the SYMMETRIC local Jacobian, the ranks of U and P , and the output-covariance rank describe different mathematical objects. They should not be interpreted as interchangeable measurements of one underlying “weight rank.” These structural results demonstrate that the quadratic adapter has a different functional form, but they do not establish that Jacobian variability, Hessian magnitude, or any measured spectral property causes the observed WikiText-2 improvement. Establishing such a causal relationship would require additional intervention-based analysis.

9.6 Computational implications

The matched rank-4 benchmark shows that the explicit quadratic operation introduces little measured practical cost in the tested configuration. Forward latency differs from LORA by approximately +0.22%, backward latency by approximately −0.27%, peak allocated VRAM increases by about 1.25 MiB, and forward throughput remains nearly unchanged.

This indicates that, in this implementation and hardware configuration, the element-wise square can be introduced without a substantial measured computational penalty. The benchmark should not be generalized to larger models, higher ranks, longer sequences, larger batches, or different accelerators without additional measurement.

9.7 Overall interpretation

Taken together, the experiments support the view that adapter functional form is an important design variable alongside rank and parameter count. Two adapters with matched trainable matrix shapes and parameter budgets can implement substantially different local transformations and can produce different empirical behaviour across tasks and optimization regimes.

Within the controlled WikiText-2 study, SYMMETRIC provides a consistent advantage over the matched LORA baseline. The AG News experiment demonstrates that this advantage does not automatically transfer to another task. The evidence therefore supports SYMMETRIC as a viable task-dependent alternative to LORA while leaving open the broader question of when quadratic activation-space interactions are most useful. Figure 9 summarizes the architectural context without implying equivalence.

Where the adaptation is applied

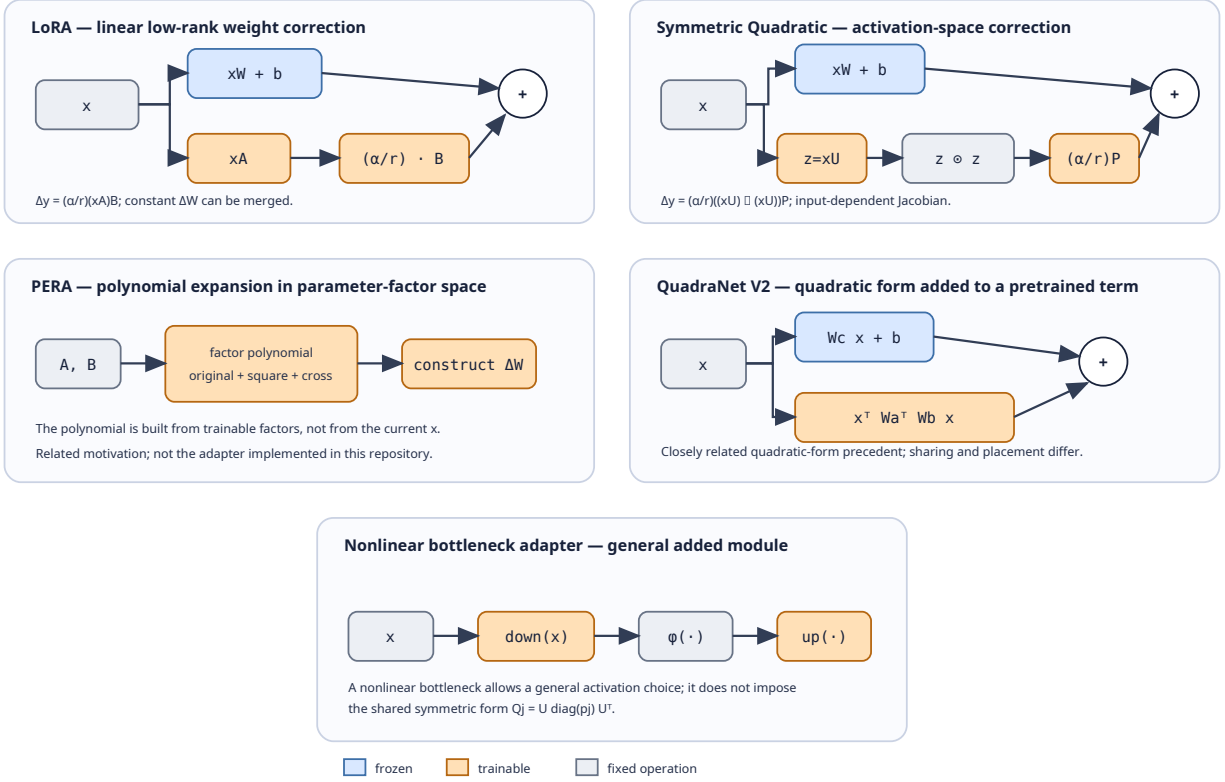


Figure 9: Architectural comparison of LORA, SYMMETRIC, PERA, QuadraNet V2, and a nonlinear bottleneck adapter. The location of a higher-order operation differs; visual similarity does not imply mathematical equivalence.

10 Limitations

This study was designed to provide controlled evidence about one specific quadratic low-rank adapter formulation rather than a complete benchmark of parameter-efficient fine-tuning. Several limitations therefore constrain the generality of the conclusions.

10.1 Model scope

The controlled experiments use GPT-2 Small. The results do not establish that the same behaviour will occur in larger GPT-2 variants, other decoder-only language models, encoder architectures, vision Transformers, or multimodal systems. Model scale and architecture may alter optimization dynamics, representation geometry, memory behaviour, and the usefulness of explicit quadratic interactions. Replication across additional model families is required before drawing broader architectural conclusions.

10.2 Task and dataset scope

The main controlled evidence comes from two datasets with different objectives: WikiText-2 causal language modelling and AG News classification. The reversed ordering demonstrates that task dependence matters, but two datasets are insufficient to characterize the conditions under which quadratic activation-space adaptation is beneficial. Additional generative, classification, reasoning, and domain-specific tasks would be required to establish broader transfer behaviour.

10.3 Experimental scope

The central WikiText-2 comparison uses three seeds, four ranks, one primary controlled adapter placement, one optimizer family, and a fixed training schedule. Three seeds provide repeated evidence but remain a small basis for broad statistical inference. The study therefore reports means, sample standard deviations, and paired differences without claiming general statistical significance over a larger population of training runs.

The principal controlled placement is zero-based block 0 `attn.c_proj`. Earlier placement and multi-layer studies provide exploratory context, but they do not constitute a fully controlled evaluation of every Transformer layer and projection.

10.4 Scaling and hyperparameters

The main rank study fixes α at 4, causing α/r to vary with rank. The constant-scale ablation examines this issue only for seed 42. The result therefore suggests that scaling alone does not explain the observed rank pattern for that seed, but it does not provide a complete multi-seed characterization of the interaction between rank and scale.

The study also does not perform an exhaustive independent hyperparameter optimization for each adapter. Alternative learning rates, schedules, initialization policies, regularization settings, or scales could alter the relative behaviour.

10.5 Mechanistic interpretation

The Jacobian and Hessian analyses establish analytical and measured structural differences between LoRA and SYMMETRIC, but they do not demonstrate that these differences cause the WikiText-2 performance gap. The adapter-only Hessian is analytically tractable. A full Transformer-block Hessian was not evaluated because it would require a separate scalar-reduction protocol and substantially greater computational complexity. More detailed intervention studies would be needed to connect local second-order structure directly to task performance.

10.6 Computational-cost scope

The computational benchmark covers one rank-4 configuration with batch size 1, sequence length 128, and one recorded hardware/software environment. The near-equal latency and small memory difference should therefore be interpreted as an implementation-specific measurement, not as a universal statement about computational efficiency. Scaling behaviour for larger models, higher ranks, longer sequences, larger batches, and different accelerators remains unmeasured.

10.7 Related-work scope

LoRA is the only related method reproduced as a direct controlled baseline. DoRA, MoRA, HiRA, LoRAN, PERA, QuadraNet V2, and nonlinear adapter methods establish mathematical and architectural context, not experimentally reproduced comparisons under the same protocol. Published results from those works are therefore not directly compared numerically with the present experiments.

The literature review also does not establish universal mathematical novelty relative to every prior quadratic neural architecture. The contribution should instead be understood as the formulation, controlled evaluation, and structural characterization of this specific activation-space quadratic adapter.

10.8 Generalization

The results establish reproducible behaviour within the tested conditions. They do not establish state-of-the-art performance, universal superiority over LoRA, broad transfer across model families, or a general causal advantage of second-order adaptation. Broader replication across larger models, additional tasks, additional placements, and independently repeated scaling conditions is required to determine how widely the observed behaviour generalizes.

11 Conclusion

SYMMETRIC is a viable quadratic activation-space adapter. Under controlled 5,000-step WikiText-2 conditions, it achieves lower mean final validation and held-out test loss than matched LoRA at ranks 1,2,4,8. AG News favours LoRA on average, so the evidence is task dependent. Rank, scaling, and duration qualify the result. Jacobian and Hessian analyses establish functional differences, while the matched benchmark finds small overhead. Adapter functional form therefore matters alongside parameter count, but broader generalization requires additional models, tasks, placements, and replications.

A Per-seed results and provenance

Per-seed WikiText-2 values are available in `results/tables/rank-confirmation-5000/rank_5000_per_seed.csv`; AG News runs are in `results/tables/downstream-ag-news/downstream_controlled_results.csv`. Aggregate tables in this paper are reconstructed from those files without cherry-picking. Figure provenance is recorded in `results/manifests/FIGURE_MANIFEST.md`.

B Historical evidence

The historical experiments preceded the final controlled protocols and were used to explore adapter formulation, training duration, placement, and distribution of a fixed parameter budget across layers. Because these campaigns were not conducted under one uniform protocol, they are reported here as exploratory evidence and are not pooled with the controlled rank-confirmation or AG News results.

B.1 Adapter-family exploration

Table 10 reports the verified three-seed best-recorded validation summaries from the 500-step screen. These exploratory runs are not a matched final benchmark.

Table 10: Historical 500-step adapter-family screen: mean best-recorded validation loss over three seeds.

Adapter	Loss	Adapter	Loss
LoRA	3.5732	Element-wise quadratic	3.6351
Signed quadratic	3.6257	Interaction	3.5581
SYMMETRIC	3.5526	Linear+Quadratic	3.5302

B.2 Longer exploratory training

Table 11 reports best-recorded validation loss at the historical rank-4 block-0 insertion point. These are minima recorded within each run and are not final-checkpoint test results.

Table 11: Historical three-seed mean best-recorded validation loss at rank 4, block 0.

Steps	LoRA	SYMMETRIC	Linear+Quadratic
1,000	3.5491	3.4770	3.4928
2,000	3.4598	3.3784	3.4360

B.3 Placement and depth exploration

A seed-42, 20-step attention scan evaluated all twelve GPT-2 blocks. SYMMETRIC recorded lower validation loss at 11 of 12 blocks; block 1 was the exception. At block 2, the largest short-run separation was 3.8243 for LoRA versus 3.8030 for SYMMETRIC. Separate attention-versus-MLP screens found nearly tied single-module MLP results, while the combined block-0 MLP branch used 15,360 parameters and therefore was not directly budget matched to the 6,144-parameter attention branches. These short screens established that placement merited further control; they do not define a stable depth ranking.

B.4 Fixed-budget multilayer exploration

Table 12 summarizes verified three-seed, 2,000-step best-recorded validation loss at matched total budgets. Distributing a fixed budget changes both placement and local rank, so these results are not merged with the later single-placement rank confirmation.

B.5 Role of the historical experiments

These exploratory campaigns served to identify promising adapter structures and experimental questions. They motivated the later controlled studies but do not carry the same evidential weight because their protocols differ in placement, duration, rank allocation, or evaluation definition. The central conclusions of this manuscript are therefore based on the controlled WikiText-2 confirmation, AG News classification, scaling, convergence, mechanism, and computational-cost experiments.

Table 12: Historical 2,000-step mean best-recorded validation loss for fixed-budget geometries.

Budget	Geometry	Local rank	LoRA	SYMMETRIC
6,144	Concentrated [2]	4	3.5039	3.3745
6,144	Two layers [0,11]	2	3.3661	3.3730
6,144	Four layers [0,3,7,11]	1	3.3199	3.3794
12,288	Concentrated [2]	8	3.5013	3.3391
12,288	Two layers [0,11]	4	3.3157	3.2390
12,288	Four layers [0,3,7,11]	2	3.2578	3.2356

C Reproducibility notes

Canonical tables are under `results/tables/`; commands and recorded software/hardware details are in `research/reproducibility/REPRODUCIBILITY.md`. Analytical Jacobian, Hessian, spectral, aggregation, and public-data tests are part of the repository suite. Raw private checkpoints are not embedded in the public manuscript.

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